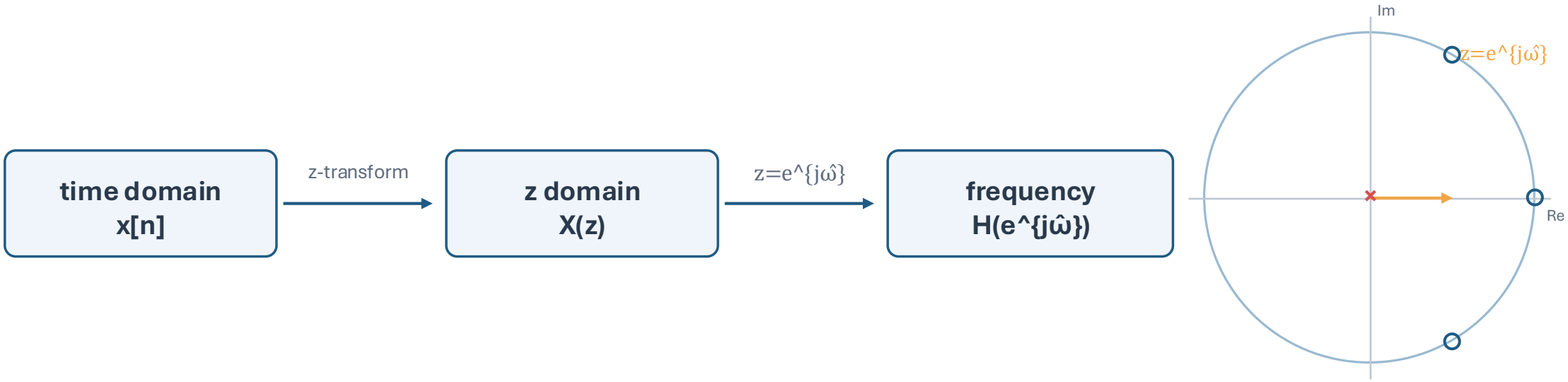
Topic 7

The z-Transform

From sequences and delays to polynomials, poles, zeros, and frequency response



Key idea. The z-transform turns time-domain filtering into algebra: delays become powers of z^{-1} and convolution becomes multiplication.

Definition of the z-transform

A sequence becomes a polynomial in z^{-1}

$$x[n] = \sum_k x[k]\,\delta[n-k]$$

$$X(z) = \sum_n x[n]\,z^{-n}$$

finite sequence $\rightleftarrows$ finite polynomial in z^{-1}

coefficients

x[0]	x[1]	x[2]	x[3]

$$X(z) = x[0] + x[1]z^{-1} + x[2]z^{-2} + x[3]z^{-3} + \cdots$$

- Each sample multiplies the matching power of z^{-1}.
- Missing powers simply have coefficient 0.

Key idea. For finite sequences, the z-transform is just a polynomial whose coefficients are the samples.

Finite and infinite z-transforms

From a polynomial to a general transform

Finite-length signal (n = 0,…,N)

$$X(z) = \sum_{n=0}^{N} x[n] \quad z^{-n}$$

A finite list of samples gives a finite polynomial in powers of z^{-1}.

General (two-sided) sequence

$$X(z) = \sum_{n=-\infty}^{\infty} x[n] \quad z^{-n}$$

The sum may contain infinitely many terms. Its convergence depends on the chosen value of z.

The z-transform maps x[n] into a function of a complex variable z.

Key idea. Finite sequences lead to polynomials; the same definition extends to infinite sequences.

Example 1: sequence ⇄ z-transform

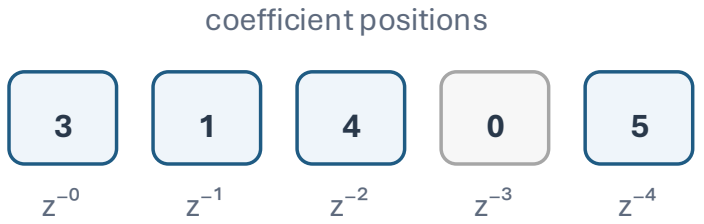
Read the coefficients of powers of z^{-1}

$$x[n] = \{\, 3, 1, 4, 0, 5 \,\} \text{ starting at } n=0$$

1. Write one term per sample: $3,\ 1z^{-1},\ 4z^{-2},\ 0z^{-3},\ 5z^{-4}$.
2. Combine the terms into a polynomial in z^{-1}.

$$X(z)=3+z^{-1}+4z^{-2}+5z^{-4}$$

3. The missing z^{-3} term means $x[3]=0$.
4. A shifted impulse $\delta[n-5]$ has z-transform z^{-5}.



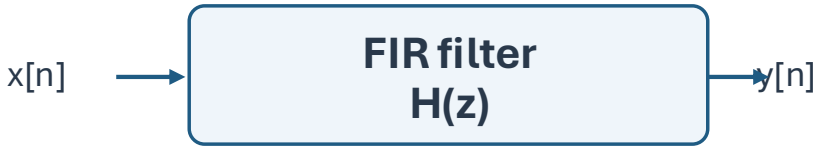
The z-transform of an FIR filter

The system function is the z-transform of h[n]

$$y[n] = b[0]x[n] + b[1]x[n-1] + \cdots + b[M]x[n-M]$$

$$h[n] = \sum_{k=0}^{M} b[k]\,\delta[n-k]$$

$$H(z) = \sum_{k=0}^{M} b[k]\,z^{-k}$$



Equivalent descriptions



Key idea. For an FIR filter, the coefficient list, impulse response, difference equation, and H(z) contain the same information.

Why the FIR system function is H(z)

Apply a complex exponential input to the FIR difference equation

$$x[n] = z^{n}$$

Substitute into the FIR equation:

$$y[n] = \sum_{k=0}^{M} b[k] \; z^{n-k}$$

$$y[n] = z^{n} \sum_{k=0}^{M} b[k] \; z^{-k}$$

$$H(z) = \sum_{k=0}^{M} b[k] \; z^{-k} = \sum_{k=0}^{M} h[k] \; z^{-k}$$

Key idea. For input z^n, the output is the same exponential multiplied by H(z).

Superposition property

The z-transform is a linear transformation

$$Z\{a\,x_1[n] + b\,x_2[n]\} = a\,X_1(z) + b\,X_2(z)$$

A concrete example:

$$x_1[n] = \delta[n], \qquad x_2[n] = \delta[n-2]$$

$$x[n] = 2x_1[n] - 3x_2[n]$$

$$X(z) = 2 - 3z^{-2}$$

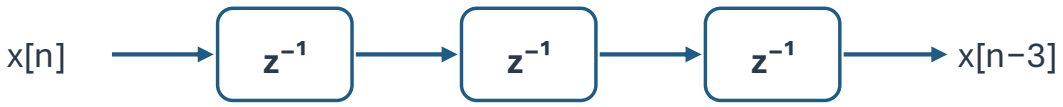
Key idea. You can transform each part of a sum independently, then combine the results.

Time shifting and the unit-delay operator

A delay becomes multiplication by z^{-1}

$$x[n-n_0] \rightleftarrows z^{-n0}\, X(z)$$

$$\delta[n-1] \rightleftarrows z^{-1}$$



Interpretation:

- one z^{-1} block = one-sample delay
- m cascaded blocks = m-sample delay
- useful in block diagrams and difference equations

Key idea. The z-transform makes delay algebraic, so block diagrams can be read as polynomials.

Worked example: delaying a sequence

Shift the entire polynomial by one power of z^{-1}

Starting sequence:

$$X(z) = 3 + z^{-1} + 4z^{-2} + z^{-3} + 5z^{-4} + 9z^{-5}$$

Delay by one sample, x[n−1]:

$$Y(z) = z^{-1}\,X(z)$$

$$Y(z) = 3z^{-1} + z^{-2} + 4z^{-3} + z^{-4} + 5z^{-5} + 9z^{-6}$$

The coefficient at n=0 is now zero; all other samples move one index later.

Key idea. A time shift changes the power of z^{-1}, not the sample values.

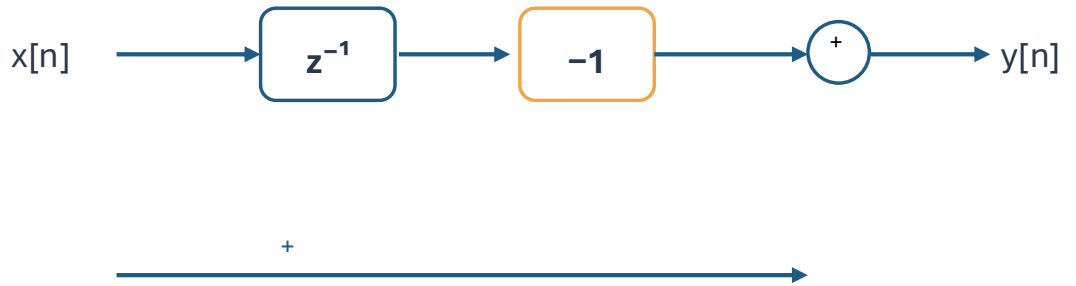
Operator notation in block diagrams

z^{-1} is a delay operator

$$y[n] = x[n] - x[n-1]$$

$$Y(z) = (1 - z^{-1})X(z)$$

$$H(z) = 1 - z^{-1}$$



This is the first-difference filter: present sample minus delayed sample.

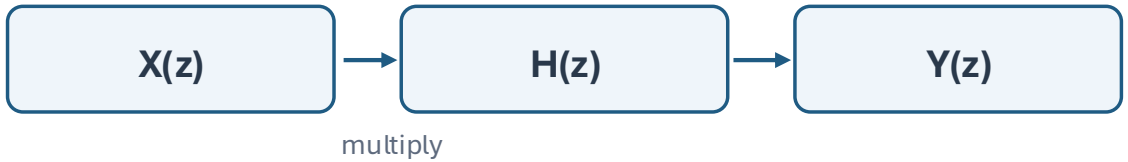
Key idea. Replacing a unit delay by z^{-1} turns block diagrams into algebraic expressions.

Convolution and the z-transform

Time-domain convolution becomes polynomial multiplication

$$y[n] = x[n] * h[n]$$

$$Y(z) = X(z)\,H(z)$$



Why this matters:

- Convolution output samples are coefficients of a polynomial product.
- Cascaded filters multiply their system functions.
- Factorising H(z) decomposes a high-order filter into smaller sections.

$$(1+2z^{-1}+z^{-2})(1-z^{-1}) = 1+z^{-1}-z^{-2}-z^{-3}$$

Key idea. The z-transform turns the convolution operation into ordinary algebra.

Why convolution becomes multiplication

Use superposition and the time-shifting property

$$y[n] = \sum_{k=0}^{M} h[k]\, x[n-k]$$

Take the z-transform of both sides:

$$Y(z) = \sum_{k=0}^{M} h[k]\, z^{-k}\, X(z)$$

X(z) does not depend on the summation index k:

$$Y(z) = X(z)\, [\sum_{k=0}^{M} h[k]\, z^{-k}]$$

$$Y(z) = X(z)\, H(z)$$

Key idea. One factor X(z) can be pulled out of the convolution sum.

Example 2: polynomial convolution

Multiply in the z-domain and read the coefficients

$$x[n]=\{1,2,1\}, \quad h[n]=\{1,-1\}$$

1. Write the z-transforms: $X(z)=1+2z^{-1}+z^{-2}$, $H(z)=1-z^{-1}$.

2. Multiply the polynomials in z^{-1}.

$$Y(z) = (1+2z^{-1}+z^{-2})(1-z^{-1})$$

$$Y(z) = 1 + z^{-1} - z^{-2} - z^{-3}$$

3. Read the output samples from the coefficients.

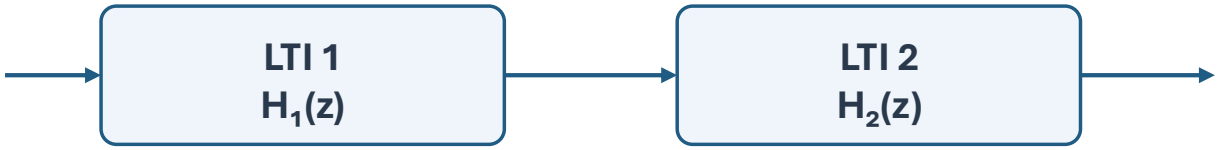
$$y[n] = \{1, 1, -1, -1\}$$



Key idea. Polynomial multiplication and finite convolution produce the same coefficient sequence.

Cascaded systems and factorisation

Products of smaller filters make larger systems



$$H_eq(z) = H_1(z)H_2(z)$$

$$h_eq[n] = h_1[n] * h_2[n]$$

Factorisation goes the other way:

$$H(z) = (1-z^{-1})(1+2z^{-1})$$

A higher-order polynomial can be implemented as a cascade of lower-order sections.

Key idea. Cascades multiply in the z-domain; factorisation turns algebra into implementation structure.

Example 2: cascade and equivalent FIR

Multiply system functions, then read coefficients

$$H_1(z)=1-z^{-1}, \quad H_2(z)=1+2z^{-1}$$

1. Overall response is the product of the two sections.

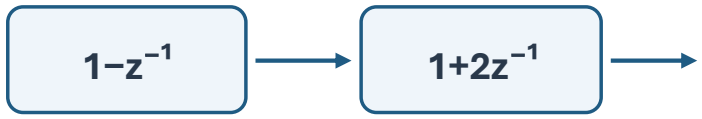
$$H_eq(z) = (1-z^{-1})(1+2z^{-1})$$

2. Expand the polynomial.

$$H_eq(z) = 1 + z^{-1} - 2z^{-2}$$

3. Read the equivalent impulse response / coefficients.

$$h_eq[n] = \{1, 1, -2\}$$



Key idea. The same result is obtained if the sections are reversed because polynomial multiplication is commutative.

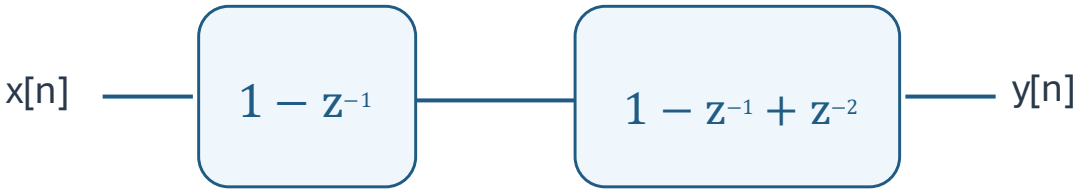
Factorisation: decompose a higher-order FIR

A zero at z=1 exposes a first-order section

$$H(z) = 1 - 2z^{-1} + 2z^{-2} - z^{-3}$$

Check H(1)=0, so $(1-z^{-1})$ is a factor.

$$H(z) = (1 - z^{-1})(1 - z^{-1} + z^{-2})$$



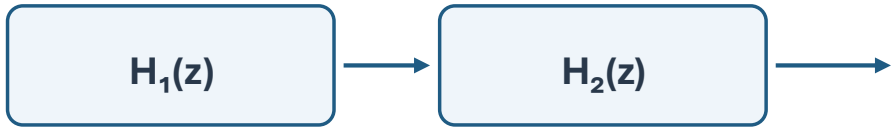
Key idea. Factoring a polynomial provides a practical cascade implementation.

Inverse filtering and deconvolution

Undoing a filter means multiplying by its reciprocal

$$H_1(z)H_2(z) = 1$$

$$H_2(z) = 1 \,/\, H_1(z)$$



overall = 1

Important consequence:

- The inverse of a finite FIR polynomial is often a rational function.
- Its power-series expansion may be infinite.
- An exact causal inverse is often IIR, not FIR.

$$1/(1-az^{-1}) = 1 + az^{-1} + a^2z^{-2} + \cdots$$

Key idea. Deconvolution is algebraically simple, but exact inverses of FIR filters are often infinite-length systems.

Inverse filtering: a useful worked example

Reciprocal system functions can produce infinite impulse responses

$$H_1(z) = 1 - a\,z^{-1}$$

$$H_2(z) = 1 \,/\, (1 - a\,z^{-1})$$

$$H_2(z) = 1 + a\,z^{-1} + a^2\,z^{-2} + a^3\,z^{-3} + \cdots$$

- The two stages multiply to 1: the overall output equals the original input.
- The causal power series converges for $|a| < 1$; an inverse is not always stable.

Key idea. The algebraic inverse of an FIR filter need not be finite length.

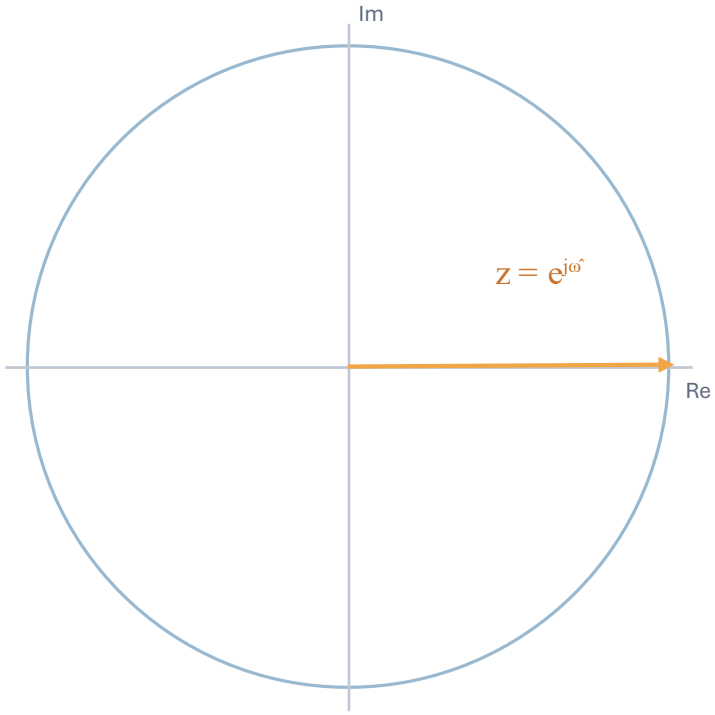
Relation between z-domain and frequency domain

Evaluate H(z) on the unit circle

$$z = e^{j\hat{\omega}}$$

$$H(e^{j\hat{\omega}}) = H(z) \mid z = e^{j\hat{\omega}}$$

- $\hat{\omega}$ is normalized radian frequency.
- As $\hat{\omega}$ moves from $-\pi$ to π, z moves once around the unit circle.
- Zeros on the unit circle cause frequency-response nulls.



Key idea. The frequency response is the trace of the z-domain system function around the unit circle.

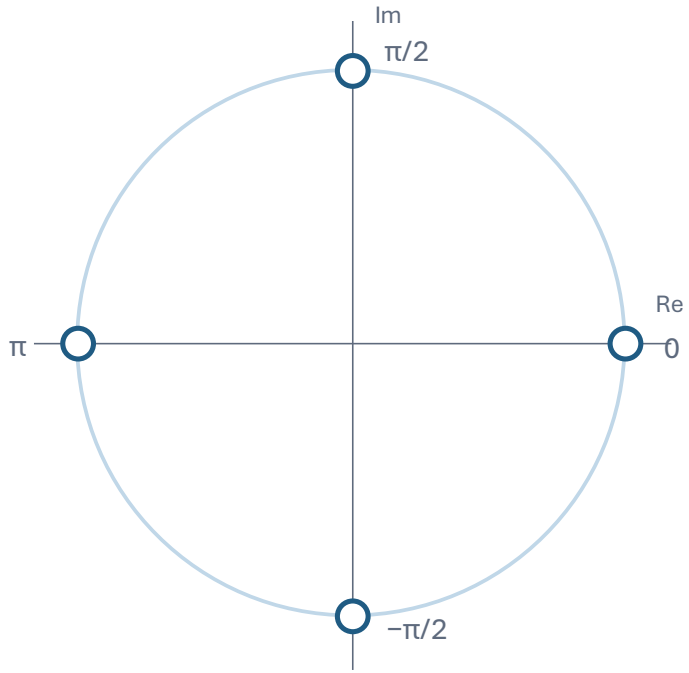
The unit circle connects z and frequency

A frequency sweep travels around the unit circle

$$z = e^{j\hat{\omega}} = \cos(\hat{\omega}) + j\sin(\hat{\omega})$$

$$H(e^{j\hat{\omega}}) = H(z) \mid_{z=e^{j\hat{\omega}}}$$

- $\hat{\omega} = 0$ corresponds to $z=1$.
- $\hat{\omega} = \pi$ corresponds to $z=-1$.
- $\hat{\omega} = \pm\pi/2$ reaches the imaginary axis.



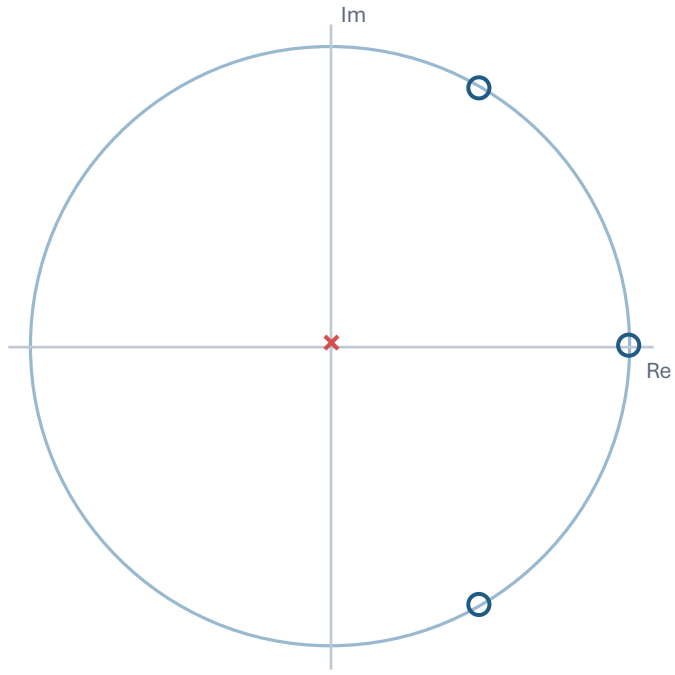
Key idea. The 2π-periodicity of $H(e^{j\hat{\omega}})$ follows from $e^{j(\hat{\omega}+2\pi)} = e^{j\hat{\omega}}$.

Poles and zeros of H(z)

Polynomial factors reveal where the response vanishes or blows up

$$H(z) = K \prod_i (1 - z_i z^{-1}) \,/\, \prod_r (1 - p_r z^{-1})$$

- Zeros z_i make $H(z_i)=0$.
- Poles p_r make $H(z)$ undefined.
- Finite FIR filters have zeros plus poles at the origin from negative powers of z.
- Zeros on the unit circle create exact frequency nulls.



Key idea. Pole-zero diagrams make the frequency-selective behaviour of a filter visible.

Pole-zero example

A third-order FIR filter with three unit-circle zeros

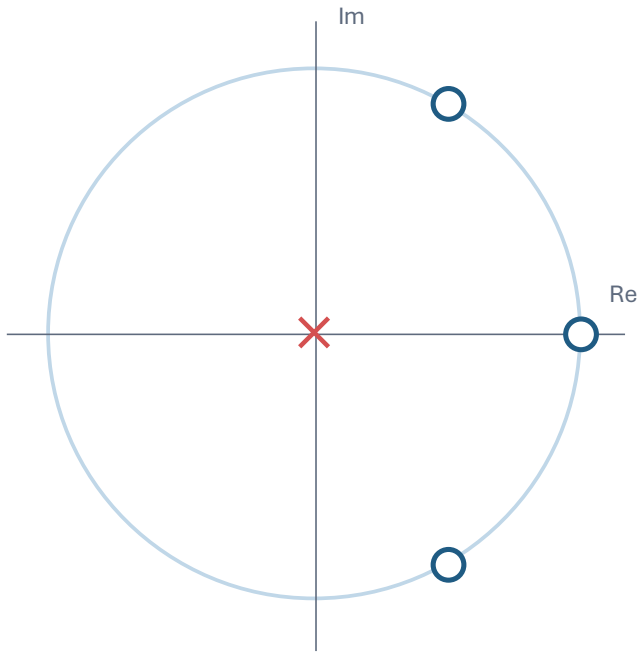
$$H(z) = 1 - 2z^{-1} + 2z^{-2} - z^{-3}$$

$$H(z) = (1 - z^{-1})(1 - e^{j\pi/3}\, z^{-1})(1 - e^{-j\pi/3}\, z^{-1})$$

Zeros:

$$z_1 = 1, \quad z_2 = e^{j\pi/3}, \quad z_3 = e^{-j\pi/3}$$

Three poles at z=0 arise from the z^{-3} factor.



Three zeros; triple pole at the origin

Key idea. Those angles correspond to three exact nulls in frequency.

Example 3: from H(z) to poles and zeros

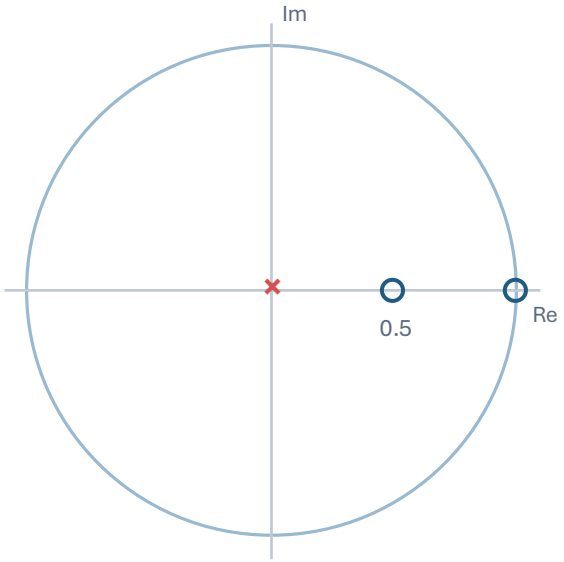
Factor the polynomial and place the roots

$$H(z)=1-1.5z^{-1}+0.5z^{-2}$$

1. Multiply by z^2 to obtain an ordinary polynomial.

$$z^2H(z)=z^2-1.5z+0.5=(z-1)(z-0.5)$$

2. Zeros are at z=1 and z=0.5.
3. The z^{-2} form introduces two poles at the origin.
4. The zero at z=1 lies on the unit circle, so DC is nulled.



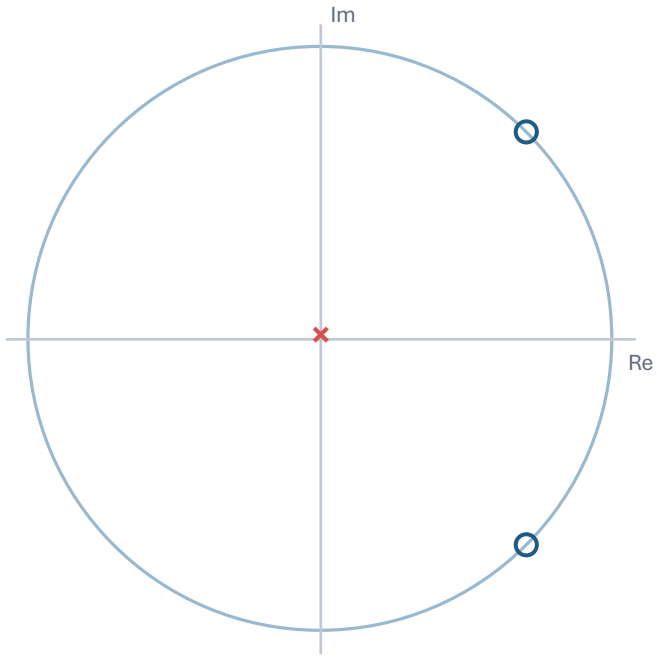
Key idea. A zero on the unit circle corresponds directly to a frequency that is completely removed.

Nulling filters

Place zeros on the unit circle at the unwanted frequency

$$H(z) = (1 - e^{j\hat{\omega}_0} z^{-1})(1 - e^{-j\hat{\omega}_0} z^{-1})$$

$$H(z) = 1 - 2\cos(\hat{\omega}_0) z^{-1} + z^{-2}$$



Why two zeros?

- A real cosine contains positive and negative frequency exponentials.
- To remove the real sinusoid, both conjugate components must be nulled.
- Real coefficients require conjugate-symmetric zeros.

Key idea. A sinusoid is suppressed when the unit-circle point at its normalized frequency coincides with a zero.

What does a zero actually suppress?

Use the exponential-input response $y[n] = H(z_0)\, z_0^n$

$$x[n] = z_0^n \quad \Rightarrow \quad y[n] = H(z_0)\, z_0^n$$

$$H(z_0) = 0 \quad \Rightarrow \quad y[n] = 0$$

For a real sinusoid:

$$\cos(\hat{\omega}_0 n) = \tfrac{1}{2}\, e^{j\hat{\omega}_0 n} + \tfrac{1}{2}\, e^{-j\hat{\omega}_0 n}$$

- Place conjugate zeros at the unit-circle frequencies $+\hat{\omega}_0$ and $-\hat{\omega}_0$.
- For a sinusoid switched on at n=0, a finite FIR may have a short transient.

Key idea. A zero at z_0 makes the response to z_0^n vanish.

Example 4: design a sinusoid-nulling FIR filter

Suppress x[n]=cos(πn/4)

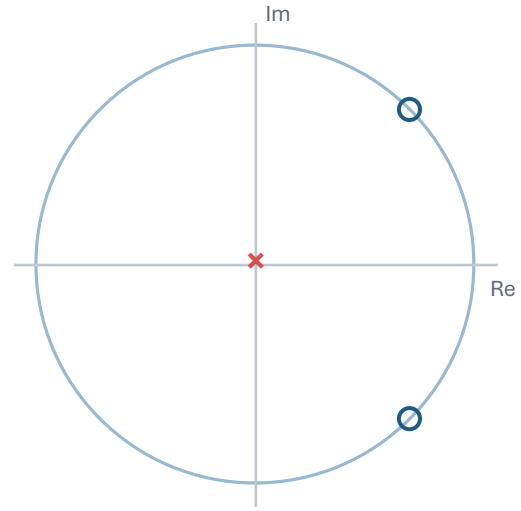
1. The cosine contains frequencies +π/4 and −π/4.
2. Place zeros at $z = e^{j\pi/4}$ and $z = e^{-j\pi/4}$.
3. Use the conjugate-zero product.

$$H(z) = (1 - e^{j\pi/4}z^{-1})(1 - e^{-j\pi/4}z^{-1})$$

4. Expand using 2cos(π/4)=√2.

$$H(z)=1-\sqrt{2}z^{-1}+z^{-2}$$

$$y[n]=x[n]-\sqrt{2}x[n-1]+x[n-2]$$



Running-sum filters and roots of unity

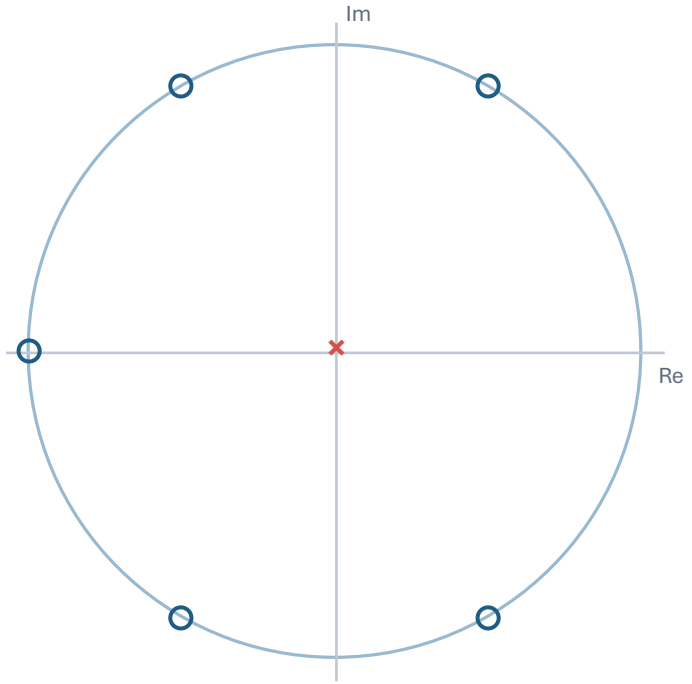
A simple FIR with regularly spaced unit-circle zeros

$$y[n] = \sum_{k=0}^{L-1} x[n-k]$$

$$H(z) = 1 + z^{-1} + \cdots + z^{-(L-1)}$$

$$H(z) = (1 - z^{-L}) / (1 - z^{-1})$$

- Zeros are the L roots of unity except z=1, which cancels.
- The missing zero at z=1 leaves strong DC response.
- The remaining zeros produce exact nulls at normalized frequencies 2πk/L.



Key idea. Running sums behave like low-pass filters with regularly spaced frequency nulls.

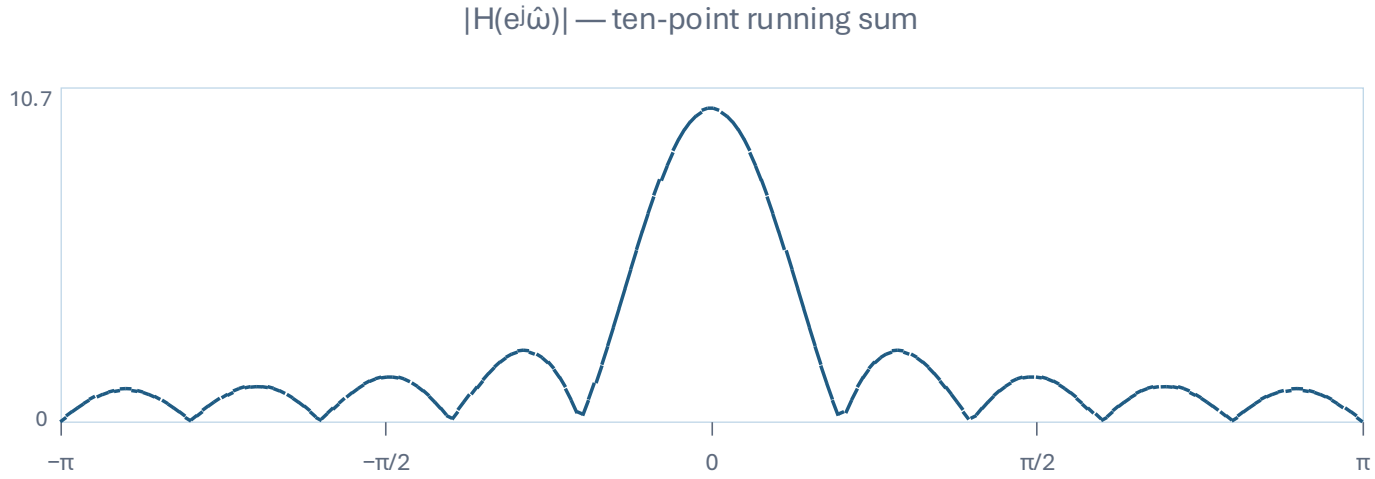
Example: 10-point running-sum FIR

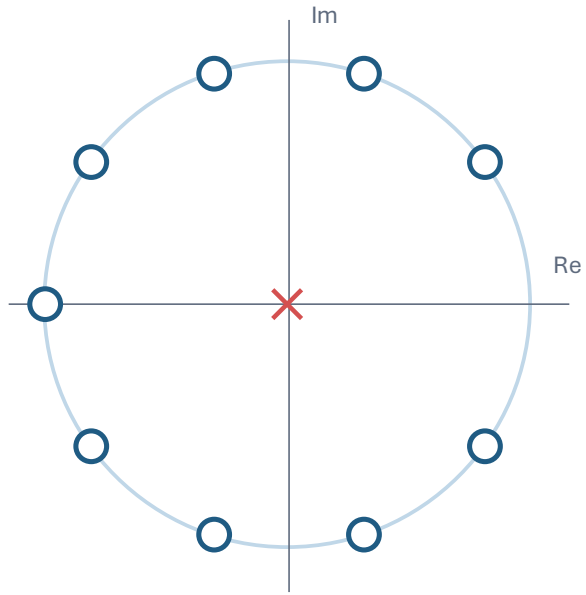
The nine non-DC roots of unity are frequency-response nulls

$$H(z) = 1 + z^{-1} + \cdots + z^{-9}$$

$$H(z) = (1 - z^{-10})/(1 - z^{-1})$$

Zeros at $\hat{\omega} = 2\pi k/10$, k=1,…,9





Nine zeros; order-nine pole at 0

Key idea. With no zero at z=1, the DC gain is H(1)=10.

From running-sum to band-pass filters

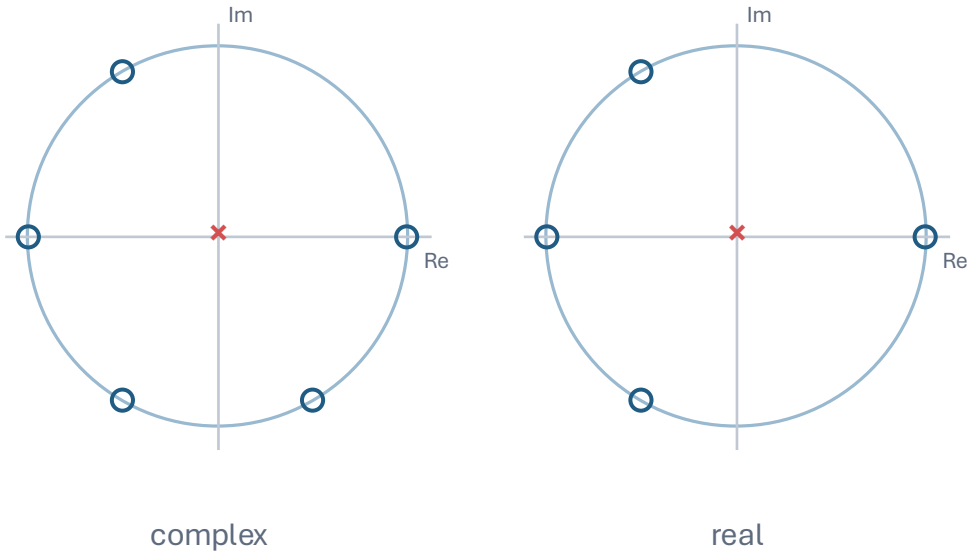
Omit a different root to move the passband

Complex band-pass:

- A running-sum-like zero pattern can be shifted.
- If one root is omitted at $+\hat{\omega}_0$, the response is largest near $+\hat{\omega}_0$.
- The coefficients are generally complex unless the zero pattern is conjugate symmetric.

Real band-pass:

- A real filter needs a matching passband at $-\hat{\omega}_0$.
- Conjugate symmetry gives real coefficients and equal magnitude lobes at $\pm\hat{\omega}_0$.



Key idea. Zero patterns on the unit circle can be shaped to pass selected frequency regions.

Complex band-pass: L = 10, k_0 = 2

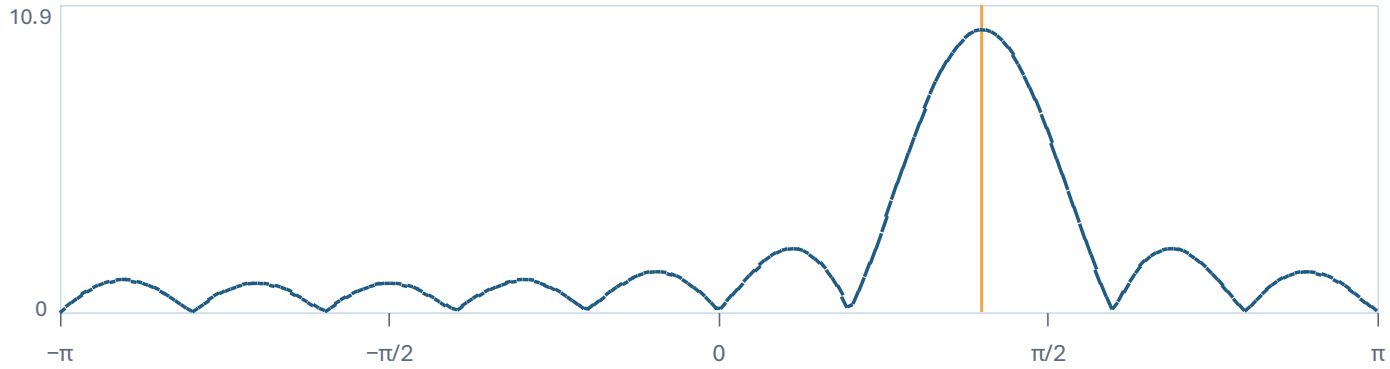
Shift the running-sum passband to the positive frequency 0.4π

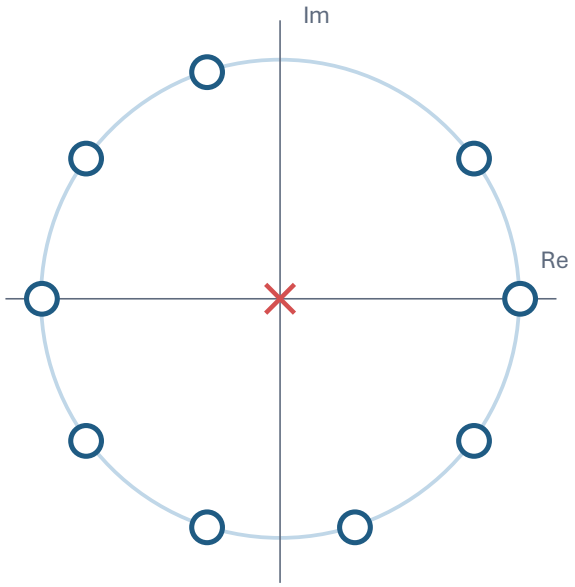
$$b[k] = e^{j(2\pi k_0/10)k}, \qquad k=0,\ldots,9$$

$$H(z) = \sum_{k=0}^{9} b[k]\, z^{-k}$$

At $\hat{\omega}_0$ = 0.4π, all ten terms add in phase.

One dominant positive-frequency passband





Omitted zero at +0.4π

Key idea. Complex coefficients allow a single passband near +0.4π.

Real band-pass: cosine-modulated coefficients

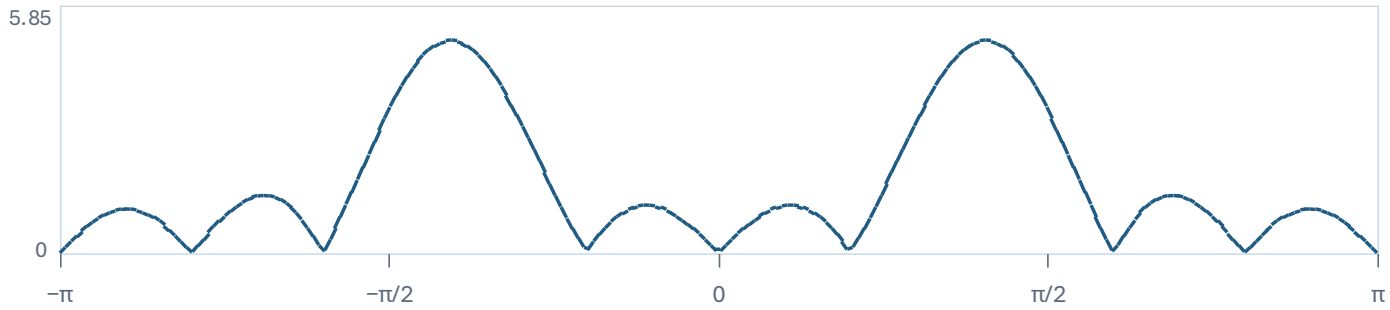
Two symmetric passbands arise at positive and negative frequencies

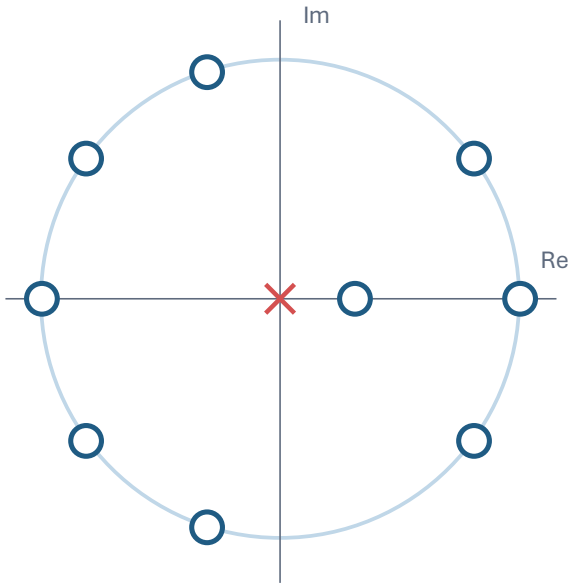
$$b[k] = \cos(2\pi\, k_0 k/10)$$

$$H(z) = \sum_{k=0}^{9} \cos(0.4\pi k)\; z^{-k}$$

The real cosine is the average of positive- and negative-frequency complex exponentials.

Equal-height passbands near ±0.4π





Conjugate-symmetric zeros

Key idea. Real coefficients require conjugate-symmetric pole-zero geometry.

Practical design: a 24-tap band-pass FIR

Windowed FIR design, corresponding to the T7 fir1 example

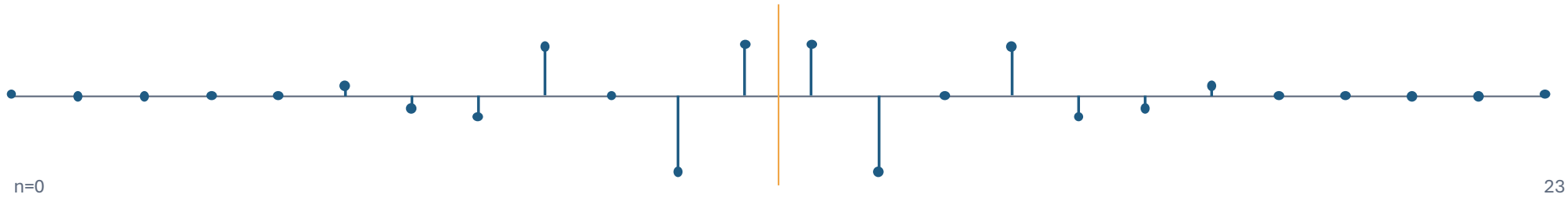
Design target (normalized radian frequency)

- Passband roughly 0.48π to 0.72π.
- Reject low- and high-frequency components.
- 24 taps ⇒ FIR order 23.

Example coefficient-design methods:

fir1 / firwin: windowed sinc design
remez: equiripple design

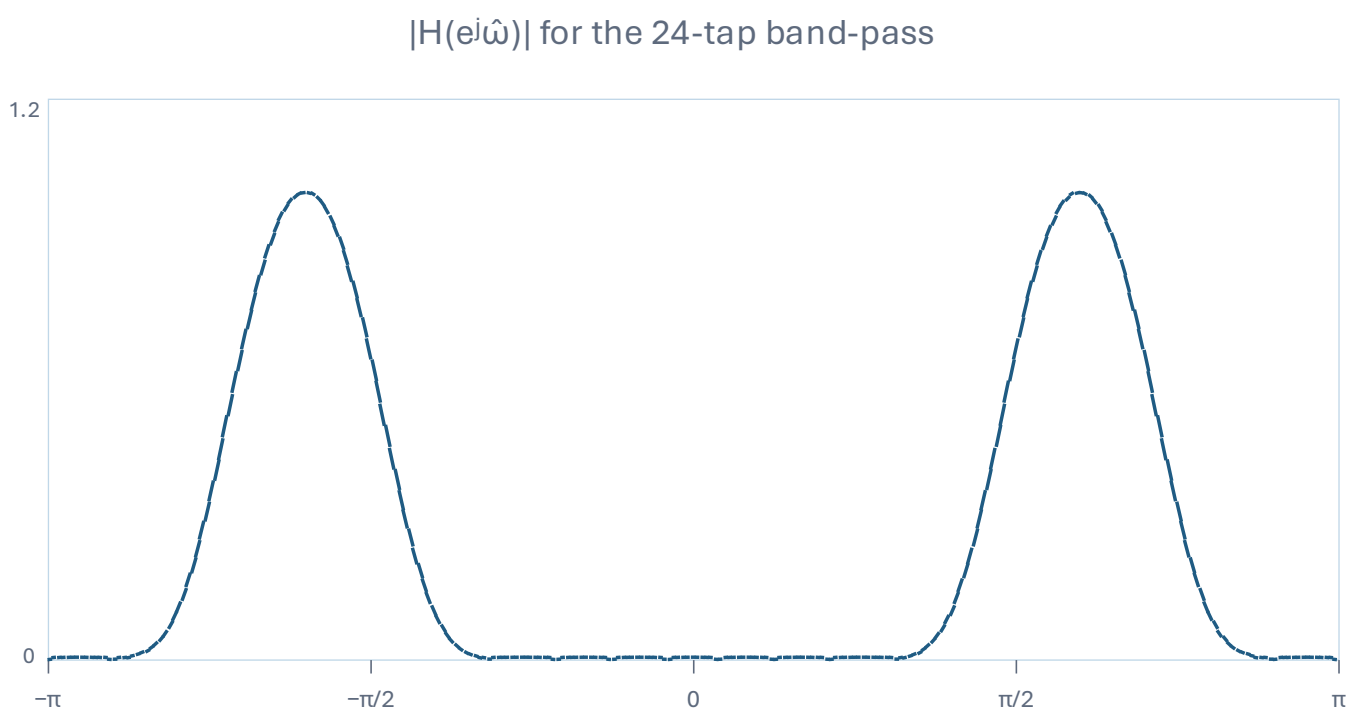
Symmetric impulse response h[n] (24 coefficients)

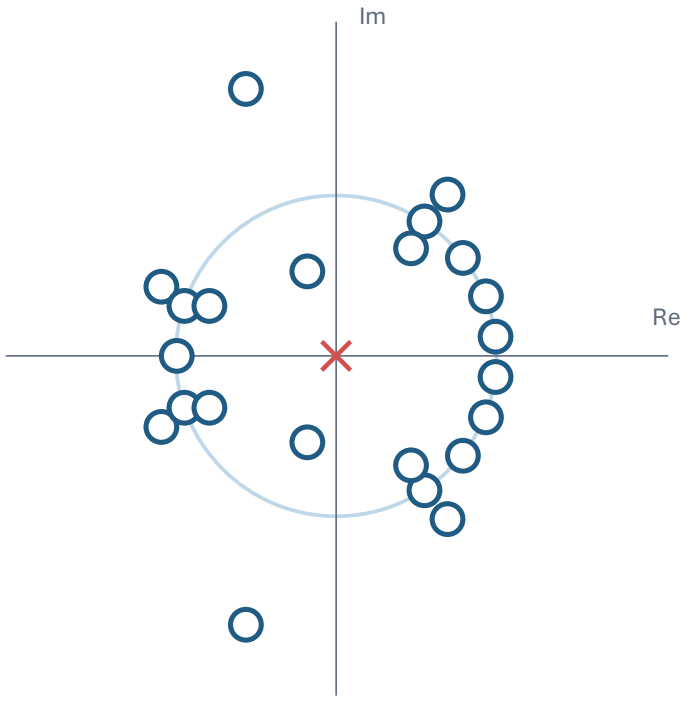


Key idea. Practical FIR methods approximate desired bands instead of placing a few zeros manually.

Practical band-pass: response and zeros

Connect the impulse response to the two-sided passband





23 zeros; order-23 pole at 0

Because the coefficients are real and symmetric, the magnitude response is symmetric in ω̂.

Key idea. The designed response is approximately zero in the stopbands and symmetric about DC.

Linear-phase FIR systems

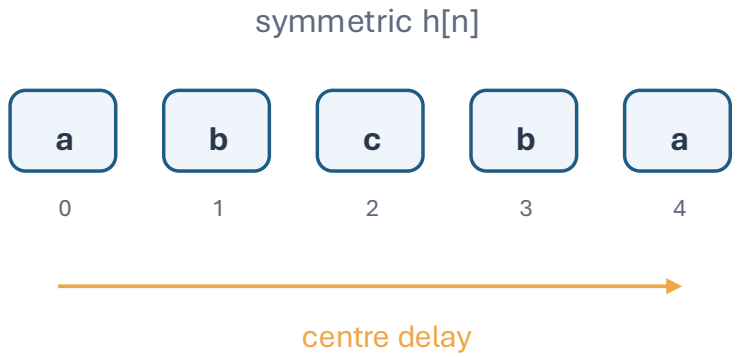
Symmetric coefficients produce a pure delay plus a real amplitude factor

$$h[n] = h[M-n]$$

$$H(e^{j\omega}) = e^{-j\omega M/2}\, A(\omega)$$

Interpretation:

- $A(\hat{\omega})$ is real, so it changes magnitude and may add π jumps.
- The exponential factor contributes a delay of M/2 samples.
- Waveform shape is preserved apart from filtering and delay.



Key idea. Linear phase is one of the main reasons FIR filters are widely used in audio and signal processing.

Derive linear phase from a symmetric 5-tap FIR

Pair equal coefficients around the central sample

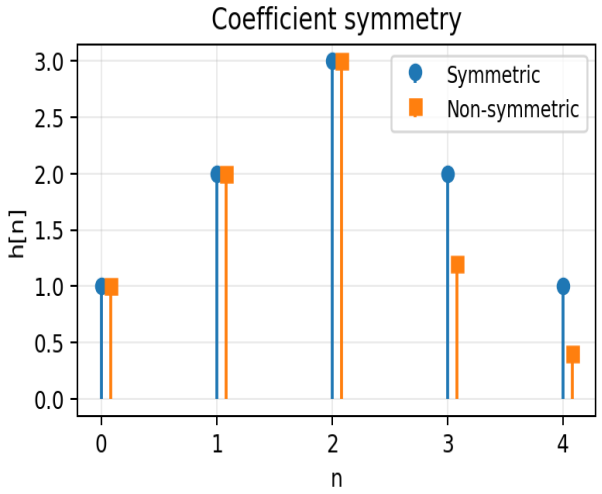
$$H(z) = b_0 + b_1 z^{-1} + b_2 z^{-2} + b_1 z^{-3} + b_0 z^{-4}$$

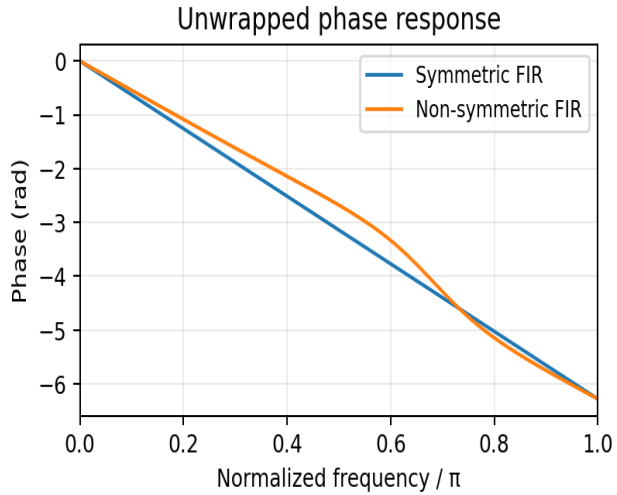
Factor out the delay of the centre tap (n = 2):

$$H(z) = z^{-2}[b_0(z^2 + z^{-2}) + b_1(z + z^{-1}) + b_2]$$

Evaluate on $z = e^{j\hat{\omega}}$ and use $e^{jk\hat{\omega}} + e^{-jk\hat{\omega}} = 2\cos(k\hat{\omega})$:

$$H(e^{j\hat{\omega}}) = e^{-j2\hat{\omega}}\,[2b_0\cos(2\hat{\omega}) + 2b_1\cos(\hat{\omega}) + b_2]$$





General case: a symmetric FIR of order M has group delay M/2 samples.

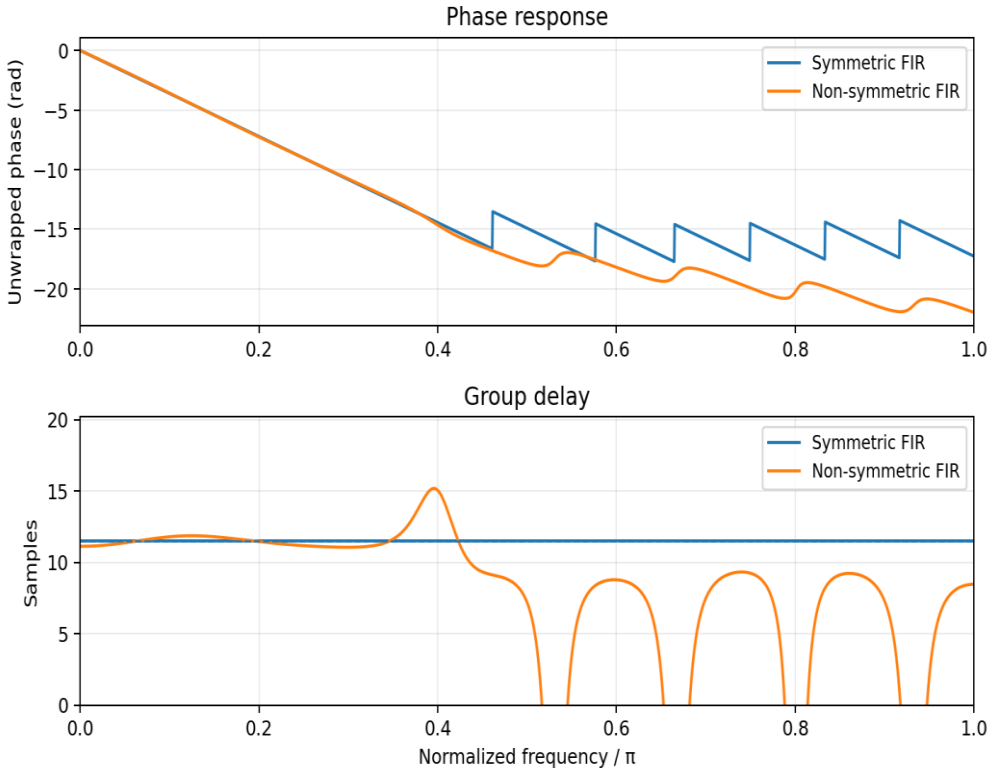
Key idea. Symmetry yields a real amplitude factor and a pure two-sample delay.

What linear phase means for a waveform

Magnitude is modified, but components share the same delay

$$H(e^{j\hat{\omega}}) = A(\hat{\omega})\, e^{-j\hat{\omega}M/2}$$

- $A(\hat{\omega})$ is real; its sign may introduce phase jumps of π.
- The linear phase term contributes constant group delay M/2.
- For the 24-tap filter, this delay equals 23/2 = 11.5 samples.



Key idea. Within the passband, a symmetric FIR does not introduce frequency-dependent delay.

Notation and takeaways

Minimum language for z-transform analysis

$x[n]$	discrete-time sequence	$H(e^{j\hat{\omega}})$	frequency response on unit circle
$X(z)$	z-transform	zero	point where H(z)=0
$H(z)$	system function / transfer function	pole	point where H(z) is undefined
z^{-1}	one-sample delay	unit circle	z=e^{jω̂}

Three core ideas

- Time-domain delays become powers of z^{-1}.
- Convolution becomes multiplication of z-domain polynomials.
- Zeros and poles explain the frequency response and filter behaviour.